

A Dual-Time Formalism for Closure Dynamics and Cosmic Evolution

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July 2026

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Abstract

This paper presents the first formalization of *dual time*: the structural distinction between electromagnetic time (t_{EM}), which is universal and governs the propagation of fields, and material time (t_{mat}), which is closure-specific and governed by entropy, mass, recursion index, and interaction structure. Building on the triadic closure framework, we introduce a conjectural dynamical relation linking these two temporal layers and refine earlier qualitative insights into eight explicit, testable conjectures. These conjectures capture entropic acceleration, mass-scaled damping, closure synchronization, the non-vanishing nature of material time, the horizon-bounded limit, and a Hawking-calibrated scaling of black-hole evolution.

Applying the formalism to celestial closures shows that dual-time dynamics reproduce the observed diversity of aging rates, lifetimes, and stability across stars, planets, degenerate matter, and black holes. The Earth’s near-unity ratio $dt_{\text{mat}}/dt_{\text{EM}}$ creates the illusion of universal time, while even small deviations from unity produce large differences on cosmic timescales. This *Earth Bias* explains why terrestrial physics appears temporally uniform and why early-universe structures observed by JWST seem anomalously mature.

A key result is the reinterpretation of Planck time as a limit of electromagnetic time rather than material time. Material-time resolution is closure-dependent and may be smaller or larger than the Planck scale. This distinction provides new perspectives on the Big Bang, inflation, and early structure formation, reframing classical singularities and superluminal expansion as cross-layer temporal mismatches rather than breakdowns of physical law.

The paper concludes with structural corollaries and a broad set of testable predictions spanning spacecraft aging, EM flux anomalies, ultrafast material response, radiation reaction, black-hole evaporation, and cosmological measurements. The dual-time architecture thereby transforms time from a single global parameter into a layered, closure-dependent dynamical quantity, offering a unified and empirically actionable temporal framework for the Triadic Cosmos.

Significance Statement

Time is treated as a single universal parameter in nearly all branches of physics, from classical mechanics and electrodynamics to quantum theory and cosmology. This paper introduces a formal dual-time architecture that challenges this assumption by separating electromagnetic time (t_{EM}), which governs the propagation of fields, from material time (t_{mat}), which governs the internal evolution of physical closures. The resulting framework provides a structural explanation for the diverse temporal behaviour observed across celestial bodies, degenerate matter, biological systems, and high-energy phenomena.

The dual-time formalism yields eight explicit conjectures that unify entropic acceleration, mass-scaled damping, closure synchronization, horizon-bounded recursion, and Hawking-calibrated black-hole evolution. These conjectures transform dual time from a conceptual insight into a mathematically tangible and empirically testable structure. The Earth’s near-unity ratio $dt_{\text{mat}}/dt_{\text{EM}}$ is shown to create the illusion of universal time, while small deviations from unity produce large differences on cosmic timescales. This *Earth Bias* reframes several long-standing astrophysical and cosmological anomalies as consequences of cross-layer temporal mismatch rather than exotic physics.

A key contribution of this work is the reinterpretation of Planck time as a limit of electromagnetic time rather than material time. Material-time resolution is closure-specific and may be smaller or larger than the Planck scale, offering new perspectives on the Big Bang, inflation, and early-universe structure formation. The formalism also provides structural explanations for EM flux anomalies, radiation-reaction pathologies, and ultrafast material response.

By proposing a layered temporal architecture and outlining a broad set of testable predictions, this paper positions dual time as a promising foundation for re-expressing time-dependent physics across scales. It offers a unified temporal perspective for the Triadic Cosmos and opens a path toward deeper theories in which matter and fields evolve on distinct but dynamically coupled clocks.

1 Introduction

Time is one of the most fundamental quantities in physics, yet nearly all physical theories treat it as a single universal parameter shared by matter, fields, observers, and the cosmos as a whole. Classical mechanics, electrodynamics, quantum theory, general relativity, and cosmology all assume that physical processes unfold on a common temporal axis. This assumption is rarely questioned, not because it is theoretically necessary, but because terrestrial physics provides no obvious reason to distinguish between different temporal layers. The Earth’s closure happens to satisfy

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \approx 1,$$

creating the illusion that material evolution and electromagnetic propagation share the same clock. This *Earth Bias* has shaped centuries of scientific intuition.

Recent empirical and theoretical developments challenge this assumption. Astrophysical systems exhibit radically different aging rates, stability profiles, and dynamical behaviour: massive stars evolve orders of magnitude faster than red dwarfs; neutron stars and white dwarfs change extremely slowly; black holes appear temporally frozen; and early-universe structures observed by JWST seem anomalously mature. These discrepancies are difficult to reconcile within a single-time framework. They suggest that matter and fields may not share the same notion of time, and that temporal evolution may be closure-dependent.

This paper develops a formal *dual-time architecture* in which electromagnetic time (t_{EM}) is universal and governs the propagation of fields, while material time (t_{mat}) is closure-specific and governed by entropy, mass, recursion index, and interaction structure. Building on the triadic closure framework, we introduce a conjectural dynamical relation linking these two temporal layers and refine earlier qualitative insights into eight explicit conjectures. These conjectures capture entropic acceleration, mass-scaled

damping, closure synchronization, the non-vanishing nature of material time, the horizon-bounded limit, and Hawking-calibrated black-hole evolution.

A central conceptual shift introduced in this work is the reinterpretation of Planck time. Traditionally viewed as the smallest meaningful temporal interval, Planck time is shown here to be a limit of electromagnetic time rather than material time. Material-time resolution is closure-dependent and may be smaller or larger than the Planck scale. This distinction provides new perspectives on the Big Bang, inflation, and early-universe structure formation, reframing classical singularities and superluminal expansion as cross-layer temporal mismatches rather than breakdowns of physical law.

The dual-time architecture developed in this paper is not merely conceptual. It yields structural corollaries that explain EM flux anomalies, radiation-reaction pathologies, ultrafast material response, and the apparent universality of terrestrial time. It also produces a broad set of testable predictions across spacecraft aging, astrophysical evolution, degenerate matter stability, black-hole evaporation, and cosmological measurements. These predictions position dual time as an empirically actionable framework capable of resolving long-standing anomalies in multiple domains.

The goal of this paper is not to claim a final theory of time, but to provide a first mathematically tangible and empirically testable formalization of dual time. The dynamical relation introduced here is provisional, and future work may refine or replace its specific functional form. What matters is the structural insight: matter and fields evolve on distinct but dynamically coupled clocks, and recognizing this distinction opens a path toward a unified temporal architecture for the Triadic Cosmos.

2 Formal Conjectures of Dual Time

This section refines the structural theorems of the previous Dual Time paper into explicit conjectures and introduces a formal dynamical relation between electromagnetic time (t_{EM}) and material time (t_{mat}). Together, these conjectures define the temporal behaviour of closures under entropy, mass, recursion index, and inter-closure interaction, and establish the limiting behaviour of material time in horizon-bounded domains.

2.1 Dual-Time Dynamics

Material time is defined as the internal temporal flow of a closure, governed by its entropic state, recursion index, and mass. Electromagnetic time is closure-transversal and universal. The interaction between these two temporal layers is captured by the following conjectural dynamical relation:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = 1 + \alpha S - \beta M_{\text{eff}} - \kappa(I) (t_{\text{mat}} - t_{\text{EM}}), \quad (1)$$

where S is the entropic footprint of the closure, M_{eff} its effective mass (including overlap contributions from neighbouring closures), and $\kappa(I)$ a diffusion coefficient determined by electromagnetic interaction strength I . Equation (1) is not a derived law but a structural ansatz that unifies the temporal behaviour described in the conjectures below.

2.2 Conjecture 1: Entropic Acceleration of Material Time

Entropy accelerates material time. As the entropic footprint S increases, the recursion index rises and the internal temporal flow of the closure increases. This behaviour is encoded in the positive entropic term $+\alpha S$ in Eq. (1).

2.3 Conjecture 2: Structural Inevitability of Aging

Any active closure participating in the triadic cycle must produce entropy. Since entropy accelerates material time, aging is structurally inevitable. No closure with nonzero entropic production can maintain a constant material-time rate.

2.4 Conjecture 3: Material Lifetime as a Monotonic Function of Entropy

The lifetime of a closure is a monotonic function of its entropic footprint. High-entropy closures evolve rapidly (e.g. massive stars), while low-entropy closures evolve slowly (e.g. degenerate matter, cryobiotic organisms). This behaviour follows directly from the entropic acceleration term in Eq. (1).

2.5 Conjecture 4: Mass-Scaled Entropic Damping

Mass damps the entropic acceleration of material time. The rate at which entropy increases recursion index is inversely proportional to mass. In Eq. (1), this appears as the negative term $-\beta M_{\text{eff}}$, where M_{eff} includes overlap contributions from neighbouring closures:

$$M_{\text{eff}} = M + \sum_{j \in \mathcal{N}} w_{ij} M_j. \quad (2)$$

This principle explains rapid stellar evolution, moderate biological aging, long-lived degenerate-matter structures, and the near-frozen temporal behaviour of microscopic organisms.

2.6 Conjecture 5: Independence of Material Time from Electromagnetic Time

Electromagnetic time does not influence aging or recursion index. Material time evolves according to closure-internal dynamics, while t_{EM} provides a universal transversal reference. The diffusion term $-\kappa(I)(t_{\text{mat}} - t_{\text{EM}})$ in Eq. (1) expresses inter-closure synchronization, not causal influence of t_{EM} on material-time acceleration.

2.7 Conjecture 6: Non-Vanishing Rate Principle

For all physically realizable active closures, the material-time rate is strictly positive:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} > 0. \quad (3)$$

Material time cannot halt; it can only be slowed arbitrarily by mass damping and reduced entropic drive.

2.8 Conjecture 7: Black-Hole Limit

In horizon-bounded closures, recursion halts and material time becomes a limiting quantity. The material-time rate approaches zero from above:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \rightarrow 0^+, \quad (4)$$

but never becomes exactly zero. Material time ceases as an emergent dynamical flow, yet remains formally defined as a limit rather than a vanishing rate. This refinement resolves the apparent discontinuity in horizon-bounded dynamics.

2.9 Conjecture 8: Hawking-Calibrated Mass Damping

Black-hole evaporation provides an empirical relation between mass and evolution rate in electromagnetic time:

$$\frac{dM}{dt_{\text{EM}}} \propto -\frac{1}{M^2}. \quad (5)$$

Within the dual-time architecture, the evaporation rate observed in t_{EM} must be expressible as a composition of internal horizon-bounded evolution in material time and the material-time rate itself:

$$\frac{dM}{dt_{\text{EM}}} = \frac{dt_{\text{mat}}}{dt_{\text{EM}}} \cdot \frac{dM}{dt_{\text{mat}}}, \quad (6)$$

Equation (6) is a structural identity linking the two temporal layers. Matching the empirical Hawking scaling (5) requires that the material-time rate decreases with mass in a manner consistent with the mass-damping term of Eq. (1). Specifically, for horizon-bounded closures,

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2}, \quad (7)$$

up to a closure-internal constant determined by horizon-scale recursion. This behaviour is precisely the limiting form of the mass-damping term $-\beta M_{\text{eff}}$ in Eq. (1) when M_{eff} is dominated by black-hole mass.

Conjecture 8 (Hawking-Calibrated Mass Damping). For horizon-bounded closures, the mass-damping term in the dual-time dynamics must asymptotically reproduce the Hawking evaporation scaling. Small black holes exhibit faster material time and rapid evaporation, while supermassive black holes approach the black-hole limit with near-frozen material time and extremely long lifetimes.

This conjecture anchors the structural mass-damping principle in empirical black-hole behaviour and provides a physical calibration for the dual-time dynamics. Black holes thus serve as canonical triadic closures in which horizon-bounded recursion, holographic encoding, and dual-time evolution coincide with observational data.

2.10 Summary

Together, Conjectures 1–8 provide a formal temporal architecture for dual time. They refine the intuitive structural theorems of the previous paper, introduce a dynamical relation between t_{mat} and t_{EM} , and establish the limiting behaviour of material time in high-mass and horizon-bounded domains. Dual time thereby becomes a mathematically tangible and empirically testable component of the Triadic Cosmos framework.

3 Celestial Closures and the Empirical Behaviour of Material Time

The conjectural dual-time dynamics of Eq. (1) imply that the effective rate of material time,

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = 1 + \alpha S - \beta M_{\text{eff}} - \kappa(I) (t_{\text{mat}} - t_{\text{EM}}),$$

must vary systematically across celestial closures as a function of entropy S , effective mass M_{eff} , horizon structure, and electromagnetic interaction. This section applies the formalism to a representative set of astrophysical and planetary bodies and shows that their observed evolution, aging behaviour, and temporal stability follow directly from Conjectures 1–8.

3.1 Large Black Hole

Supermassive black holes possess extreme mass and horizon-bounded recursion. Their effective mass M_{eff} is maximal, while internal entropic drive S is bounded by holographic capacity. The mass-damping term $-\beta M_{\text{eff}}$ dominates, driving

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \rightarrow 0^+.$$

Material time approaches the black-hole limit (Conjecture 7): internal evolution is effectively frozen, and all observable behaviour is encoded at the horizon. Combined with Conjecture 8, this explains the extremely slow Hawking evaporation of supermassive black holes and their near-eternal stability.

3.2 Small Black Hole

Stellar-mass black holes exhibit lower M_{eff} than supermassive black holes but retain horizon-bounded recursion. Their material-time rate is still strongly damped,

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \ll 1,$$

yet remains significantly higher than in the supermassive case. Conjecture 8 implies

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2},$$

so small black holes exhibit faster material time and therefore faster Hawking evaporation. This matches the empirical scaling of black-hole lifetimes and the rapid evaporation of primordial or low-mass black holes.

3.3 Massive Star

Large, luminous stars (e.g. O- and B-type) possess high entropic footprint S due to intense fusion, turbulent convection, and radiative output, combined with substantial mass M . The entropic acceleration term $+\alpha S$ is large, while $-\beta M$ provides partial damping. The net effect is a high material-time rate:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \gg 1,$$

leading to rapid aging, short lifetimes, and fast structural evolution. This behaviour matches observed main-sequence lifetimes of massive stars (millions of years).

3.4 Low-Mass Star

Small stars (e.g. K- and M-type red dwarfs) have lower entropic footprint and lower fusion rates, combined with moderate mass. The entropic term $+\alpha S$ is reduced, while $-\beta M$ still provides damping. The resulting material-time rate is close to unity or slightly above:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \gtrsim 1,$$

yielding slow aging and extremely long lifetimes (hundreds of billions of years). This behaviour is consistent with the empirical stability of red dwarfs.

3.5 Neutron Star

Neutron stars exhibit extreme mass density and low entropic generativity. Their effective mass M_{eff} is very large, while internal entropy production S is limited by degenerate matter. The mass-damping term dominates:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \ll 1,$$

but remains strictly positive (Conjecture 6). Neutron stars therefore evolve extremely slowly, with long spin-down and cooling timescales, matching observed longevity and temporal stability.

3.6 White Dwarf

White dwarfs possess high density and moderate mass, with low entropic activity after fusion ceases. Their S is small, M_{eff} moderate, yielding

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} < 1,$$

but not as suppressed as in neutron stars. They cool and crystallize over billions of years, consistent with a slowed but non-frozen material time.

3.7 Brown Dwarf and Giant Planet

Brown dwarfs and gas giants (e.g. Jupiter) occupy an intermediate regime: substantial mass, limited fusion or none, and moderate entropic activity from atmospheric dynamics and internal heat. The balance

$$1 + \alpha S - \beta M_{\text{eff}}$$

yields a material-time rate slightly below unity:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \lesssim 1.$$

Their internal evolution (cooling, contraction, atmospheric reorganization) is slow, matching the observed long-term stability of giant planets and substellar objects.

3.8 Rocky Planet (Earth)

Terrestrial planets such as Earth have lower mass than gas giants but significant entropic activity: tectonics, atmospheric circulation, chemical cycling, and biological processes. Here $+\alpha S$ and $-\beta M$ are comparable, yielding

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \approx 1,$$

with modest acceleration in high-entropy subsystems (biosphere, crust) and slower evolution in low-entropy domains (deep mantle, core). This explains the coexistence of rapid biological aging and slow geological change within the same planetary closure.

3.9 Moon

Low-mass, geologically inactive bodies such as the Moon exhibit minimal entropic drive and small mass. With S very low and M_{eff} modest, the material-time rate is close to unity but with extremely slow structural change:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \approx 1,$$

and effectively negligible aging on human timescales. This matches the empirical stability of lunar geology and surface features.

3.10 Summary

Across large black holes, small black holes, massive and low-mass stars, neutron stars, white dwarfs, brown dwarfs, giant planets, rocky planets, and moons, the conjectural dual-time dynamics reproduce the observed spectrum of aging rates, lifetimes, and temporal stability. High-entropy, moderate-mass closures age rapidly; high-mass, low-entropy closures evolve slowly; horizon-bounded closures approach the black-hole limit with near-frozen material time. Conjecture 8 further anchors the behaviour of black

holes in empirical Hawking evaporation, providing a unified temporal interpretation of celestial evolution within the Triadic Cosmos framework.

4 Structural Corollaries and Illustrative Consequences of Dual Time

The formal conjectures developed in previous sections allow several intuitive observations to be expressed as structural corollaries. These corollaries illustrate how dual-time dynamics manifest in biological aging, electromagnetic flux, particle-field interaction, ultrafast material response, and the apparent universality of time in terrestrial physics.

4.1 Corollary 1: Environmental Mass Modulates Aging

Conjectures 1 and 4 imply that the effective material-time rate of a closure depends not only on its internal entropy S and mass M , but also on the mass of its surrounding environment through M_{eff} . The ratio

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = 1 + \alpha S - \beta M_{\text{eff}}$$

therefore predicts that aging slows down in high-mass environments and accelerates in low-mass environments. This explains why biological aging is slightly slower on Earth than in low-mass space environments, extremely slow near massive objects, longer in large organisms, and nearly frozen in microscopic closures where mass dominates the entropic ratio. Environmental mass does not replace entropy; it modulates its temporal effect.

4.2 Corollary 2: EM Flux as a Cross-Closure Temporal Comparison

Conjectures 1, 5, and 8 imply that electromagnetic radiation provides a direct window into closure-specific material-time rates. EM-time is universal, while material-time is closure-dependent. A closure with a high recursion index emits EM events into the global t_{EM} clock at a rate determined by its own t_{mat} . An observer with a different material-time rate detects these events through their own t_{mat} , producing the appearance of anomalously high EM flux:

$$\text{Observed EM flux} \propto \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \right)_{\text{source}} / \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \right)_{\text{observer}}.$$

This corollary provides a complementary interpretation of the extraordinary luminosity of pulsars, magnetars, gamma-ray bursts, and early-universe phase transitions: their material time runs faster than ours, while EM-time does not.

4.3 Corollary 3: Dual Time Resolves Radiation-Reaction Pathologies

Radiation reaction in classical electrodynamics produces pre-acceleration and runaway solutions when particle dynamics and field dynamics are forced to share a single global time. Conjecture 5 implies that particle evolution occurs in t_{mat} while field propagation occurs in t_{EM} . When these two clocks are separated, the pathological solutions become time-inconsistent and disappear. Dual time therefore provides a structural resolution of radiation-reaction anomalies without modifying the underlying electrodynamics.

4.4 Corollary 4: Ultrafast Pump–Probe Dynamics as Clock Mismatch

Ultrafast pump–probe experiments reveal delayed response, asymmetric peaks, long relaxation tails, and memory-like behaviour in electronic systems. Conjectures 1 and 5 imply that pump pulses evolve in t_{EM} while electronic relaxation evolves in t_{mat} . A simple mapping between these two clocks reproduces the observed features without additional dynamical terms. Dual time therefore provides a natural interpretation of ultrafast material response as a cross-clock interaction.

4.5 Corollary 5: Dual Time as a Structural Upgrade to Time-Dependent Physics

Modern physics assumes a single global time for particles, fields, materials, and observers. Conjectures 1–8 imply that this assumption is a closure-specific artefact rather than a cosmic principle. Re-expressing time-dependent equations with two clocks— t_{EM} for fields and t_{mat} for matter—removes several long-standing anomalies, simplifies the interpretation of ultrafast and high-energy processes, and provides a structural foundation for cross-closure temporal comparison. Dual time does not replace existing physics; it reorganizes it into a more coherent temporal architecture.

4.6 Corollary 6: The Earth Bias and the Illusion of Universal Time

The Earth’s closure satisfies

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \approx 1,$$

placing terrestrial physics in a narrow regime where entropic acceleration and mass damping nearly cancel. This creates the illusion of a universal temporal flow. On cosmic scales, even small deviations from unity in this ratio produce enormous differences in evolution, stability, and observable behaviour. The apparent universality of time is therefore a local artefact of the terrestrial closure rather than a cosmic principle.

4.7 Summary

These corollaries demonstrate that dual-time dynamics provide intuitive, structurally grounded explanations for phenomena ranging from biological aging and EM flux to radiation reaction, ultrafast material response, and the apparent universality of terrestrial time. They illustrate how Conjectures 1–8 reorganize diverse empirical behaviour into a coherent temporal architecture and highlight the explanatory power of dual time across domains.

5 Planck Time and the Dual-Time Architecture

The Planck time,

$$t_{\text{P}} = \sqrt{\frac{\hbar G}{c^5}} \approx 5.39 \times 10^{-44} \text{ s},$$

is traditionally interpreted as the smallest meaningful unit of time in physics. Its definition depends exclusively on the constants c , $\hbar$, and G , which encode the propagation of electromagnetic fields, quantum action, and gravitational coupling. Crucially, none of these constants describe closure-internal dynamics, entropy production, recursion index, or mass-damping. Within the dual-time framework, this implies that Planck time is a limit of electromagnetic time rather than a universal limit of temporal evolution.

5.1 Planck Time as an Electromagnetic-Time Limit

Electromagnetic time t_{EM} is closure-transversal and governs the propagation of photons, field quanta, and holographic information. The Planck time therefore represents the smallest resolvable interval of t_{EM} in which electromagnetic processes retain physical meaning:

$$\Delta t_{\text{EM}}^{\text{min}} = t_{\text{P}}.$$

This limit applies uniformly across all closures, independent of their internal structure. Planck time is thus a fundamental pixel of the electromagnetic clock.

5.2 Absence of a Universal Material-Time Minimum

Material time t_{mat} , by contrast, is closure-specific and governed by the dual-time dynamics of Eq. (1):

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = 1 + \alpha S - \beta M_{\text{eff}} - \kappa(I) (t_{\text{mat}} - t_{\text{EM}}).$$

Because this rate depends on entropy, effective mass, recursion index, and closure interaction, material time does not inherit the Planck limit. Instead, the smallest resolvable interval of material time is closure-dependent:

$$\Delta t_{\text{mat}}^{\text{min}} = \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \right) \Delta t_{\text{EM}}^{\text{min}} = \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \right) t_{\text{P}}.$$

Closures with high entropic acceleration ($+\alpha S$) may exhibit material-time intervals smaller than t_{P} , while closures dominated by mass damping ($-\beta M_{\text{eff}}$) may exhibit material-time intervals vastly larger than t_{P} . Material time therefore has no universal minimum.

5.3 Implications for Closure Dynamics

Black holes. In horizon-bounded closures, the material-time rate approaches zero from above (Conjecture 7):

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \rightarrow 0^+.$$

Material time becomes arbitrarily slow, while electromagnetic time retains its Planck-scale resolution. This explains the effective freezing of internal evolution and the persistence of Hawking evaporation as an EM-time process (Conjecture 8).

Early-universe closures. High entropic drive and low mass damping imply

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \gg 1,$$

allowing material-time intervals smaller than t_{P} . This provides a natural interpretation of rapid early-universe evolution and the apparent over-maturity of high-redshift galaxies.

Degenerate matter. Neutron stars and white dwarfs exhibit extremely low entropic generativity and high effective mass, yielding

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \ll 1.$$

Material-time intervals become much larger than t_{P} , explaining their extreme stability and slow evolution.

Terrestrial physics. The Earth's closure satisfies

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \approx 1,$$

placing material-time intervals near the Planck scale and creating the illusion of a universal temporal resolution. This is a manifestation of the Earth Bias.

5.4 Conceptual Consequences

The dual-time interpretation of Planck time has several fundamental implications:

- Planck time is not the smallest possible interval in which material changes can occur.
- Material-time resolution varies across closures and may be smaller or larger than t_P .
- The Planck scale governs electromagnetic processes, not closure-internal evolution.
- The apparent universality of Planck time is a consequence of Earth-like closures where $t_{\text{mat}} \approx t_{\text{EM}}$.
- Closure-specific temporal resolution provides a new lens for interpreting early-universe dynamics, black-hole evolution, and degenerate matter stability.

5.5 Summary

Planck time is a fundamental limit of electromagnetic time, not material time. Within the dual-time architecture, material-time resolution is closure-dependent and may diverge significantly from the Planck scale. This distinction resolves several conceptual tensions in cosmology and high-energy physics and reinforces the structural separation between t_{mat} and t_{EM} as independent temporal layers.

6 Dual Time, Planck Time, and the Early Universe

The dual-time architecture provides a new perspective on the Big Bang, inflation, and the role of Planck time in early-universe dynamics. In standard cosmology, the Planck time

$$t_P = \sqrt{\frac{\hbar G}{c^5}}$$

is interpreted as the smallest meaningful temporal interval, below which classical notions of space-time cease to apply. Within the dual-time framework, this interpretation is revised: Planck time is a limit of electromagnetic time (t_{EM}), not of material time (t_{mat}). This distinction has profound implications for the earliest stages of cosmic evolution.

6.1 Planck Time as an EM-Time Limit in the Early Universe

Electromagnetic time governs the propagation of photons, field quanta, and holographic information. Its smallest resolvable interval is the Planck time:

$$\Delta t_{\text{EM}}^{\text{min}} = t_P.$$

This limit applies uniformly across all closures, including the primordial closure of the early universe. However, material time is closure-specific and governed by the dual-time dynamics:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = 1 + \alpha S - \beta M_{\text{eff}} - \kappa(I) (t_{\text{mat}} - t_{\text{EM}}).$$

In the early universe, entropy production S and recursion index are extremely high, while effective mass M_{eff} is initially low. This implies

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \gg 1,$$

allowing material-time intervals smaller than the Planck time:

$$\Delta t_{\text{mat}}^{\text{min}} = \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \right) t_{\text{P}} \ll t_{\text{P}}.$$

Material changes can therefore occur at rates faster than the Planck-scale resolution of electromagnetic time.

6.2 Implications for the Big Bang

The Big Bang is traditionally viewed as a singularity where spacetime curvature diverges and classical physics breaks down. In the dual-time framework, the primordial closure possesses an extremely high entropic footprint and recursion index, yielding a material-time rate far above unity. This implies:

- The earliest material evolution may have proceeded at rates faster than the Planck time allows for electromagnetic processes.
- The apparent singularity may reflect the mismatch between t_{mat} and t_{EM} , rather than a breakdown of physical law.
- The Big Bang can be interpreted as a closure with extremely rapid material-time evolution embedded within an EM-time layer that retains its Planck-scale limit.

This perspective reframes the singularity problem as a temporal-layer mismatch rather than a geometric divergence.

6.3 Implications for Inflation

Inflation requires an extremely rapid expansion of spacetime within a tiny EM-time interval. In standard cosmology, this expansion appears superluminal and requires a scalar field with special properties. Within the dual-time architecture, inflation emerges naturally from the closure-specific behaviour of material time:

- High entropic drive ($+\alpha S$) and low mass damping ($-\beta M_{\text{eff}}$) imply $\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \gg 1$.
- Material-time evolution proceeds far faster than EM-time evolution.
- Processes that appear instantaneous or superluminal in t_{EM} may be slow and continuous in t_{mat} .

Inflation is therefore reinterpreted as a phase where the primordial closure evolves rapidly in material time while electromagnetic time remains constrained by the Planck limit. This removes the need for superluminal expansion and reframes inflation as a cross-clock phenomenon.

6.4 Early-Universe Structure Formation

High-redshift galaxies observed by JWST appear “too evolved” relative to standard cosmic time. Dual time provides a natural explanation:

- Early closures have $\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \gg 1$.
- Material-time evolution proceeds much faster than EM-time evolution.
- Structures that appear anomalously mature in t_{EM} may be consistent with rapid evolution in t_{mat} .

This resolves the tension between early-universe observations and single-time cosmology.

6.5 Summary

Planck time is a fundamental limit of electromagnetic time, not material time. In the early universe, material-time evolution may proceed at rates far exceeding the Planck-scale resolution of t_{EM} . This distinction provides new interpretations of the Big Bang, inflation, and early structure formation, reframing several classical cosmological puzzles as consequences of cross-layer temporal mismatch rather than breakdowns of physical law.

7 Testable Predictions

The dual-time formalism developed in this paper yields a broad set of empirically testable predictions across spacecraft dynamics, astrophysical evolution, planetary processes, biological aging, electromagnetic phenomena, and cosmology. These predictions follow directly from Conjectures 1–8 and provide a concrete experimental and observational program for evaluating the dual-time architecture.

7.1 Spacecraft and Interstellar Probes

Prediction 1: Reproducibility of the Voyager Effect. A low-entropy, low-recursion probe leaving the heliosphere will exhibit measurably slower material degradation than identical probes operating within the terrestrial closure, while maintaining perfect electromagnetic coherence. (Conjectures 1, 4, 5.)

Prediction 2: Entropy-Dependent Aging in Deep Space. High-entropy probes (warm electronics, high dynamical complexity) will age faster outside the heliosphere than low-entropy probes, even under identical external conditions. (Conjectures 1, 4.)

Prediction 3: Closure-Boundary Drift Near the Heliopause. Spacecraft approaching the heliopause will exhibit small but measurable deviations in inertial behaviour consistent with a changing ratio $dt_{\text{mat}}/dt_{\text{EM}}$. (Conjectures 4, 5.)

7.2 Astrophysical Evolution

Prediction 4: Early-Universe Over-Evolution. High-redshift galaxies observed by JWST will systematically appear “too evolved” relative to standard cosmic-time expectations due to elevated material-time rates in early closures. (Conjectures 1, 5, Earth Bias.)

Prediction 5: Ubiquity of Early Supermassive Black Holes. Supermassive black holes at $z > 7$ must exist generically, as early closures with high recursion index produce accelerated material-time evolution and rapid mass assembly. (Conjectures 1, 4, 8.)

Prediction 6: Extreme Stability of Neutron Stars. Neutron stars will exhibit cooling curves, spin-down rates, and magnetic decay timescales significantly slower than predicted by single-time models. (Conjectures 4, 6.)

7.3 Planetary and Biological Processes

Prediction 7: Aging Rate Depends on Environmental Mass. Biological aging will be measurably slower in higher-mass environments (e.g. gas giants) and measurably faster in low-mass environments (e.g. orbital habitats). (Conjectures 1, 4.)

Prediction 8: Near-Frozen Material Time in Microscopic Organisms. Cryptobiotic organisms (tardigrades, rotifers, nematodes) will exhibit material-time rates significantly below unity, explaining their extreme longevity and suspended animation. (Conjectures 1, 4, 6.)

Prediction 9: Closure-Specific Radiocarbon Decay. Radiocarbon decay rates will remain constant within a closure but exhibit detectable differences in interstellar or extra-solar biological material. (Conjectures 1, 5.)

7.4 Electromagnetic Phenomena

Prediction 10: EM Flux Anomalies as Clock-Mismatch Effects. Systems with high recursion index will appear overluminous relative to standard expectations, producing higher EM event rates per unit of observer material-time. (Conjectures 1, 5.)

Prediction 11: Ultrafast Pump–Probe Asymmetry from Dual Clocks. Ultrafast electronic response in materials will exhibit asymmetric peaks, delayed onset, and long relaxation tails that can be reproduced by mapping between t_{EM} and t_{mat} . (Conjectures 1, 5.)

Prediction 12: Resolution of Radiation-Reaction Pathologies. Pre-acceleration and runaway solutions in classical electrodynamics will disappear when particle dynamics are expressed in t_{mat} and field propagation in t_{EM} . (Conjecture 5.)

7.5 Black-Hole Dynamics

Prediction 13: Hawking Evaporation Rate Matches Dual-Time Scaling. Small black holes will evaporate significantly faster than large ones, with evaporation rates obeying

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2},$$

as required by Conjecture 8.

Prediction 14: Horizon-Bounded Recursion-Halt Signatures. Black-hole ringdown, extremal-surface saturation, and the absence of interior degrees of freedom will follow directly from the recursion-halt limit $dt_{\text{mat}}/dt_{\text{EM}} \rightarrow 0^+$. (Conjecture 7.)

7.6 Cosmology

Prediction 15: Closure-Dependent Hubble Measurements. Local measurements of the Hubble constant will differ systematically from CMB-derived values due to closure-specific material-time rates. (Conjectures 5, Earth Bias.)

Prediction 16: CMB Anomalies Reflect Recursion Variation. Large-scale CMB anomalies (hemispherical asymmetry, cold spot, quadrupole alignment) will correspond to closure-dependent recursion differences in the early universe. (Conjectures 1, 5.)

Prediction 17: Apparent Universality of Time Only in Earth-Like Closures. Closures satisfying $dt_{\text{mat}}/dt_{\text{EM}} \approx 1$ will exhibit the illusion of universal time, while closures with even small deviations from unity will show radically different aging, evolution, and EM flux behaviour. (Earth Bias, Conjectures 1–8.)

7.7 Summary

These predictions demonstrate that dual-time dynamics are not merely conceptual but empirically actionable. They define a broad experimental and observational program spanning spacecraft engineering, astrophysics, planetary science, condensed-matter physics, and cosmology. The dual-time architecture is

therefore testable, falsifiable, and capable of producing precise quantitative deviations from single-time models.

8 Open Questions and Limitations

The dual-time architecture presented in this paper is a first attempt at formalizing the relationship between material time and electromagnetic time. While the conjectures and dynamical relations developed here provide a coherent structural framework, several open questions remain. The effective formulas used in this proposal should therefore be understood as provisional and subject to refinement as empirical data, theoretical constraints, and mathematical consistency evolve.

8.1 Provisional Nature of the Dynamical Ansatz

Equation (1) is a structural ansatz rather than a derived law. Its form captures the interplay between entropy, mass, recursion index, and closure interaction, but alternative formulations may reproduce the same qualitative behaviour. The coefficients α , β , and $\kappa(I)$ are not yet grounded in a deeper theory and may require reinterpretation or replacement in future work. The present formalism should therefore be viewed as a conceptual scaffold rather than a final expression of dual-time dynamics.

8.2 Uncertainty in the Entropic Term

The entropic acceleration term $+\alpha S$ captures the intuitive relationship between entropy production and internal temporal flow, but the precise functional dependence on S remains unknown. Whether entropy should enter linearly, logarithmically, or through a more complex recursion-index mapping is an open question. High-entropy astrophysical closures may provide empirical constraints on this term.

8.3 Ambiguity in Effective Mass

The definition of effective mass M_{eff} incorporates overlap contributions from neighbouring closures, but the weighting factors w_{ij} are not yet derived from first principles. The mass-damping term $-\beta M_{\text{eff}}$ may require modification to account for relativistic effects, horizon geometry, or nonlocal mass distributions. The Hawking-calibrated scaling in Conjecture 8 provides a strong empirical anchor, but the general form of mass damping outside horizon-bounded systems remains uncertain.

8.4 Diffusion Term and Closure Interaction

The diffusion term $-\kappa(I)(t_{\text{mat}} - t_{\text{EM}})$ models inter-closure synchronization, but the functional dependence of κ on electromagnetic interaction strength I is not yet established. Different closures may require different interaction kernels, and the present linear form may be insufficient for strongly coupled systems or high-energy environments.

8.5 Material-Time Rate in Microscopic Closures

Conjecture 6 asserts that the material-time rate is strictly positive for all active closures, but the behaviour of $dt_{\text{mat}}/dt_{\text{EM}}$ in extremely small or extremely cold systems remains unclear. Microscopic closures may require a more detailed treatment of quantum recursion, degeneracy, or entropic suppression.

8.6 Cosmological Boundary Conditions

The dual-time interpretation of early-universe evolution, CMB anomalies, and Hubble tension depends on closure-specific recursion rates in primordial environments. These rates are not yet constrained by direct observation, and alternative cosmological models may require adjustments to the dual-time mapping between t_{mat} and t_{EM} .

8.7 Need for a Deeper Underlying Theory

The present formalism provides a coherent phenomenological structure, but a deeper theoretical foundation is needed. Whether dual time emerges from quantum gravity, holographic recursion, field-theoretic layering, or a more fundamental triadic mechanism remains an open question. The conjectures developed here should therefore be viewed as a first step toward a more complete theory.

8.8 Summary

The dual-time architecture offers a unified temporal interpretation of closure dynamics, but its mathematical form is provisional. The conjectures and dynamical relations presented here are intended as a starting point for further refinement, empirical testing, and theoretical development. Future work may modify, extend, or replace the present formulas while preserving the underlying structural principles of dual time.

9 Conclusion

This paper presents a first formalization of dual time: the distinction between the closure-internal temporal flow t_{mat} and the universal electromagnetic time t_{EM} . Building on the structural insights of the previous material-time paper, we introduced a conjectural dynamical relation linking entropy, mass, recursion index, and closure interaction. The resulting architecture provides a unified temporal framework for interpreting the evolution of celestial bodies, the behaviour of matter and fields, and the apparent universality of terrestrial time.

The eight conjectures developed here refine the earlier qualitative theorems into explicit, testable statements. They capture entropic acceleration, mass-scaled damping, closure synchronization, the non-vanishing nature of material time, the horizon-bounded limit, and the Hawking-calibrated behaviour of black holes. Together, these conjectures define a coherent temporal structure in which different closures evolve at different material-time rates while sharing a common electromagnetic clock.

The application of this formalism to celestial closures demonstrates that dual-time dynamics reproduce the observed diversity of aging rates, lifetimes, and stability across stars, planets, degenerate matter, and black holes. The Earth Bias highlights that the apparent universality of time is a local artefact of the terrestrial closure, not a cosmic principle. Even small deviations in the ratio $dt_{\text{mat}}/dt_{\text{EM}}$ produce large differences on cosmic timescales, offering new perspectives on early-universe evolution, high-redshift anomalies, and the stability of compact objects.

The structural corollaries show that dual time provides intuitive explanations for phenomena ranging from biological aging and EM flux anomalies to radiation reaction and ultrafast material response. The testable predictions outline a broad experimental and observational program spanning spacecraft engineering, astrophysics, condensed-matter physics, and cosmology. Dual time is therefore not only conceptually coherent but empirically actionable.

At the same time, the dynamical relation proposed here is provisional. The precise functional dependence of entropy, mass, and interaction terms may differ from the forms introduced in this paper. The present formalism should be viewed as a first attempt at capturing the temporal structure implied by triadic closure dynamics. Future work may refine, extend, or replace these formulas while preserving the underlying principles of dual time.

In summary, this paper transforms dual time from a conceptual insight into a mathematically tangible and empirically testable framework. It provides a foundation for reinterpreting time-dependent processes across scales and opens a path toward a deeper theory in which material time and electromagnetic time coexist as structurally distinct yet dynamically coupled temporal layers.

A Dual Time as an Independent Theoretical Contribution

Although the dual-time architecture is developed within the broader context of the Triadic Cosmos, its conceptual and mathematical structure stands on its own. The distinction between electromagnetic time (t_{EM}) and material time (t_{mat}) does not depend on the triadic unification programme, nor on holographic layering, nor on the specific interpretation of closure recursion. Dual time is a structurally independent proposal that can be evaluated, tested, and applied without reference to the full Triadic Cosmos framework.

Several aspects of the present formalization highlight this independence:

- **Dual time introduces a new temporal architecture** that reorganizes time-dependent physics across scales, regardless of the underlying unification mechanism.
- **The dynamical relation between t_{mat} and t_{EM}** is self-contained and empirically testable. Its validity does not rely on triadic recursion, holography, or graviton positioning.
- **The reinterpretation of Planck time** as an EM-time limit rather than a universal temporal bound is conceptually independent and compatible with multiple theoretical frameworks, including quantum gravity and semiclassical cosmology.
- **The early-universe implications**—including the Big Bang as a temporal-layer mismatch and inflation as accelerated material-time evolution—do not require the full Triadic Cosmos to be meaningful.
- **The testable predictions** spanning spacecraft aging, EM flux anomalies, ultrafast material response, radiation reaction, black-hole evaporation, and cosmological measurements arise directly from dual time and not from other triadic components.
- **Dual time resolves classical anomalies** (e.g. radiation reaction, early-universe over-evolution, black-hole temporal freezing) through its own internal logic, independent of unification.

Dual time therefore occupies a dual role: it is a natural component of the Triadic Cosmos, where it integrates seamlessly with closure recursion, holographic encoding, and mass- scaled dynamics; but it is also a standalone theoretical contribution that can be adopted, tested, and developed independently. This independence strengthens the scientific value of the present work and ensures that the dual-time architecture remains robust even if other elements of the Triadic Cosmos are revised or replaced.

B Alternative Forms of the Dual-Time Ansatz

The dynamical relation introduced in Eq. (1),

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = 1 + \alpha S - \beta M_{\text{eff}} - \kappa(I) (t_{\text{mat}} - t_{\text{EM}}),$$

is the simplest structural form capable of expressing the temporal behaviour implied by Conjectures 1–8. It is linear, additive, and first-order in the relevant quantities. This simplicity is intentional: the goal of the present paper is to formalize dual time and establish its empirical and conceptual consequences, not to determine the unique or final functional form of the dynamics. Several alternative formulations are possible and may be explored in future work.

B.1 Multiplicative Forms

In the present ansatz, entropy, mass, and closure interaction contribute additively to the material-time rate. A multiplicative structure is equally plausible:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = (1 + \alpha S) f(M_{\text{eff}}, I),$$

where f is a damping or modulation factor. This form naturally expresses the idea that entropic acceleration may be scaled by mass or interaction strength rather than simply offset by them.

B.2 Exponentially Damped Forms

Mass damping may also enter through an exponential factor:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = e^{-\beta M_{\text{eff}}} (1 + \alpha S),$$

which captures the intuition that large effective mass suppresses material-time evolution more strongly than a linear term. This form may be relevant for degenerate matter or horizon-bounded closures.

B.3 Nonlinear Interaction Terms

The diffusion term $-\kappa(I)(t_{\text{mat}} - t_{\text{EM}})$ expresses linear synchronization between closures. Nonlinear alternatives include

$$-\kappa(I) (t_{\text{mat}} - t_{\text{EM}})^n,$$

with $n > 1$ producing stronger synchronization when closures diverge and weaker synchronization when they are nearly aligned. Such forms may be appropriate for strongly coupled plasma environments or early-universe closures.

B.4 Recursion-Index Coupling

Material-time evolution may depend explicitly on recursion index R :

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = g(R) (1 + \alpha S) - h(R) M_{\text{eff}},$$

where g and h encode how recursion amplifies or suppresses entropic and mass-driven effects. This form may be relevant for closures with rapidly changing internal structure.

B.5 Higher-Order or Mixed Forms

More general ansätze may combine additive, multiplicative, and nonlinear terms:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = F(S, M_{\text{eff}}, I, R)$$

with F an arbitrary smooth function satisfying the structural constraints of Conjectures 1–8. Such forms may arise naturally in future toy models or in deeper theoretical developments linking dual time to holographic recursion or quantum gravity.

B.6 Summary

The dual-time ansatz used in the main text is a minimal structural expression of the temporal behaviour implied by the conjectures. It is not unique, nor intended as a final form. Alternative multiplicative, exponential, nonlinear, or recursion-dependent forms are compatible with the dual-time architecture and may be explored in future work. The present paper focuses on establishing the conceptual and empirical foundations of dual time; more detailed dynamical refinements are deferred to subsequent studies.

C Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

`https://github.com/triadic-cosmos`

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